Aleksandar Dubljevic

416-312-8161 | Toronto | adubljev@uwaterloo.ca | alexdubljevic.com | LinkedIn | GitHub

Education

University of Waterloo & Wilfrid Laurier University Sept. 2023 – May 2028
Bachelor of Computer Science and Bachelor of Business Administration Double Degree *Waterloo, ON*
Relevant Coursework: Data Structures and Algorithms | Awards: Pearl Sullivan Hack the North Scholarship

Experience

Cloud Engineer Jan. 2025 – April 2025
Canadian Imperial Bank of Commerce *Toronto, ON*

- Deployed and tested Azure resource configurations (Databricks, VMs, Storage) for enterprise applications across dev/test/prod environments, completing **63 Jira tickets** with zero post-deployment rollback incidents as part of an Agile Scrum team
- Automated Azure resource deployments by executing **Terraform** scripts through **Azure DevOps** pipelines, reducing manual provisioning time by **40%** while ensuring version-controlled, consistent infrastructure across environments
- Resolved a **SEV2** outage impacting critical banking services by tracing network flows with **Azure Network Watcher** to identify misconfigured firewall rules, collaborating with senior engineers to restore connectivity for thousands of users within 2 hours
- Recognized as a **top-performing intern** and admitted into CIBC Spark, a selective program for high-achieving interns based on quickly mastering development workflows and independently contributing to production infrastructure by week 3

Software Engineer - Dashboard Lead Sept. 2023 – Present
University of Waterloo Formula Electric *Waterloo, ON*

- Led the development of an internal **Grafana** dashboard for live telemetry, enabling team members to monitor real-time performance from over **150 data sources**, eliminating the need to restart the vehicle to view sensor data
- Enhanced sensor data transfer rate by over **500%** by designing and implementing a wireless data pipeline
- Ensured timely development and deployment of team software by utilizing SDLC and Agile methodologies for effective project management, leading to increased team velocity and a **35%** reduction in work-in-progress tasks

Projects

Dime Defender | **Hack the North 2024 Finalist** — *AWS, Svelte, Voiceflow, GenAI* Sept. 2024

- Voted **1st place** among **800+ participants** at Hack the North 2024 by developing a Chrome extension that helps users reduce impulsive spending through AI-driven insights powered by **OpenAI API**
- Architected a scalable **AWS** serverless backend (Lambda, API Gateway) that supported over **75** concurrent users, achieving **sub-500ms** response times through API request optimizations like pagination and payload compression
- Led a team of 4 developers to deliver a production-ready MVP in 36 hours, coordinating development across frontend, backend, and AI components while using **GitHub** for version control

MpoxMap | **Ignition Hacks 2024 Winner** — *TypeScript, React, Next.js, HTML/CSS* August 2024

- Developed MpoxMap, a web application using **Next.js 14** and **React** that monitors global Mpox outbreaks using **3D data visualizations**, displaying real-time and historical data, keeping users informed on the latest outbreaks
- Implemented API caching optimizations to reduce redundant calls to TheNewsAPI and Mapbox API, improving page load times by **200%**, resulting in a smoother user experience and reduced API costs
- Deployed to web with Vercel serverless architecture, allowing for seamless updates to data sources

GameWave | *Java, PostgreSQL, Replit* March 2023 – June 2023

- Constructed a full-stack mock video game marketplace using Java Swing, with features such as login/signup, leaving reviews and searching and filtering for games
- Implemented a content-based filtering recommendation algorithm that analyzed user preferences to suggest games, in combination with a **PostgreSQL** database used to efficiently store and retrieve user data
- Integrated secure user authentication with **SHA-256 hashing** to safeguard sensitive user info through data masking

UWFE Dashboard | *Python, CAN, Figma, FreeRTOS, Linux* Jan. 2024 – June 2024

- Designed and developed a custom driver dashboard for the University of Waterloo Formula Electric team using Python and Figma, deployed to Linux, enhancing vehicle data visibility for drivers during competitions
- Architected a multi-threaded data processing pipeline to handle **6,000+** CAN bus messages per second, implementing thread-safe queues and efficient data structures that maintained **sub-50ms** UI update latency under peak loads
- Deployed and tested dashboard software on a standalone Beaglebone Black running **Debian Linux**

Technical Skills

Languages: Java, Python, C/C++, JavaScript, TypeScript, HTML/CSS, SQL, Svelte, Shell Script, Ruby, Go, Rust
Frameworks: React, Node.js, Next.js, FreeRTOS, CI/CD, AWS, Azure, Kubernetes, Ruby on Rails, Agile, SDLC
Developer Tools: Git, Docker, Tableau, Atlassian Suite, Power BI, MS Office, Terraform, Cursor, Claude Code, GenAI
Case Competitions: Snap Inc. Xlerate (Finalist), BDO Future Leaders, Parachute Consulting